

Ritvik Sharma

New York City, NY

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Education

Columbia University

Aug 2025 - Present

Master of Science in Computer Science - AI & NLP Track

SRM University – [Silver Medalist Amongst 700+ Students](#)

June 2019 - May 2023

Bachelor of Technology in Computer Science with Specialization in AI-ML

CGPA: 9.86/10

Technical Skills

Languages: Python, C/C++, JavaScript, MySQL

Databases: AWS DynamoDB, MongoDB, SQL Server, Oracle

Developer Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Kafka, Linux

Technologies: AWS(Route53, CloudFront, DataSync, ECS, EC2, SageMaker, OpenSearch, S3, DynamoDB, Lambda), Apache Tomcat, Django, NodeJS, Selenium, gRPC, NLP, BERT, Machine Learning & NLP

Work Experience

Barclays, UK

July 2023 - July 2025

SDE-II & DevOps

Pune, India

- **Saving \$15,000 Monthly:** Innovated utilities ensuring optimal AWS Lambda Usage. Received **Barclays-Wide Recognition** for the same.
- **96% Performance Improvement** in Glue Jobs pertaining to CRUD operations in AWS DynamoDB dealing with 5 Million+ Records (**12 hours to 27 minutes**) by optimally using Spark, Dynamic Frames & Batch Write.
- **75% Reduction in Error-Resolution Times:** Developed & Designed a complex Error-Routing Utility integrating Python-based Lambda, IBM Netcool, SNS Topic, Subscription Filters on Logs across all domains of the platform.
- **Implemented No-Downtime Deployments:** Innovatively integrated Blue-Green & Canary Based deployments in the existing system. Received **Barclays-Wide Recognition** for the same.
- **Optimized CI-CD Pipelines with 63% higher success rates:** Simplifying deployment Configs and Innovating Smart Automations. Received **Barclays-Wide Recognition** for the same.
- **Launching Barclays' Major Cloud Platform:** Successfully Launched & Led the development & designing of the Integrating layer between Onboarding, Rewards, Payments and other journeys for our first credit card **ArrowCard** on the Platform.

Projects

VELVÉT - Luxury E-Commerce Website

[Link to Live Demo](#)

- Built a fully functional luxury e-commerce site with React, Gatsby, AWS Lambda, Route53, CloudFront, DynamoDB.
- Integrated ML-based semantic search by training a Custom BERT Model and hosting on AWS SageMaker. Used AWS OpenSearch for fast product discovery.
- Implemented a secure and high quality AWS backend using IAM Roles, Basic DDoS Prevention and implemented Selenium-based unit + integration tests to ensure Production-Grade Quality.

Inter-University Hackathon Winner: Mercury

[Link to Web/Video Demo](#)

- A Chatbot built using AI/ML & NLP techniques for finding food order details from a sentence such as I want **2 Veg Margherita Pizzas** with **Olives** from **Dominos**.
- Designed and Implemented the backend of the Mobile app made with Python, Flutter and using ALBERT (LLM for NLP) and CRF (Probabilistic Prediction) and Speech To Text.

HyperSafety: Workplace COVID Safety App

[Link to Web/Video Demo](#)

- An app for Flagging & Identifying Employees without Masks to keep the Workplace Safe.
- Designed and Implemented the backend of the Mobile app where I worked on Face Detection and Recognition along with MySQL DB and NodeJS Backend.

Publications & Awards

- Sharma, Ritvik (April 2025). "VectorQuest: Faster and Better Open-Domain Question-Answering System" **Published in Springer Nature Switzerland AG** for the innovation leading to **50 times Faster Prediction**.
- [Inter-University Hackathon Winner](#) for Building an AI-Powered Chatbot for ordering food. [Check Out the Demo Here](#)